

EAST DELTA UNIVERSITY



Title

FOODGENOME AI

**A Calibrated Multi-Backbone Food Recognition System
with Retrieval-Augmented Nutritional Reasoning**

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FoodGenome AI: A Calibrated Multi-Backbone Food Recognition System with Retrieval-Augmented Nutritional Reasoning

An end-to-end system that recognises 101 food categories from a single photograph, reports how confident it is in a statistically meaningful way, refuses to answer when the photograph is not food, and grounds every nutritional statement it makes in the United States Department of Agriculture’s published food-composition data.

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Abstract

This project proposes FoodGenome AI, an end-to-end food recognition and nutritional reasoning system built on the Food-101 dataset of 101,000 photographs across 101 categories. Rather than training a single classifier, the system constructs a *frozen feature bank* by caching the embeddings of three complementary pretrained vision backbones — SigLIP-SO400M, DINOv2-L and EVA-02-L — which reduces every downstream experiment from hours to seconds and makes rigorous ablation affordable on a single laptop. Lightweight probe classifiers trained on this bank establish a strong baseline, after which EVA-02-L is fully fine-tuned at 448×448 resolution using layer-wise learning-rate decay, mixup, CutMix and an exponential moving average of weights. The final prediction is an ensemble with test-time augmentation, evaluated once on a test split that is held untouched throughout development.

The central argument of this proposal is that classification accuracy alone is an insufficient goal for a nutrition application. A confidently wrong calorie estimate is worse than no estimate at all. FoodGenome AI therefore treats *reliability* as a first-class deliverable: model confidence is calibrated, conformal prediction produces prediction sets with a guaranteed coverage rate, and an out-of-distribution detector rejects photographs that are not food. Predictions are explained with Grad-CAM and attention maps, and portion size is estimated from monocular depth. Nutritional answers are produced by a purpose-built retrieval-augmented generation pipeline — hybrid sparse and dense search, reciprocal rank fusion, cross-encoder reranking, corrective retrieval, explicit grounding and a graph-structured knowledge base derived from USDA SR Legacy — so that every figure the system reports is traceable to a source record. The trained models are exported to ONNX and published on the Hugging Face Hub, served through a containerised FastAPI backend, and consumed by a Next.js web application. The target is 96% top-1 accuracy on Food-101 with calibrated, auditable and explainable predictions.

Introduction

Dietary self-monitoring is one of the few interventions with consistent evidence behind it across weight management, diabetes control and cardiovascular health. Its weakness is not efficacy but adherence. Conventional food logging requires a person to identify a dish, search a database, choose among near-identical entries and estimate a portion — perhaps a dozen interactions for a single meal, repeated several times a day. Adherence collapses within weeks, and the tool stops producing value precisely when sustained use would matter most.

A photograph is a far more natural interface. The user points a camera at a plate; the system does the identification, the database lookup and the portion estimate. This reframes the problem as fine-grained visual classification followed by structured knowledge retrieval, and both halves are now tractable. Vision transformers pretrained on web-scale corpora transfer remarkably well to food imagery, and retrieval-augmented generation allows a language model to answer nutritional questions from an authoritative database instead of from memorised parameters.

Food classification is nonetheless a genuinely difficult fine-grained problem. The Food-101 categories include pairs that differ only in ingredients invisible from above — spaghetti bolognese against spaghetti carbonara, filet mignon against prime rib, beef carpaccio against beef tartare. Intra-class variation is extreme: the dataset is deliberately built from unretouched user photographs, so lighting, plating, camera angle and cuisine-specific presentation vary enormously within every class. The training split is also known to contain a degree of label noise, which the original authors left in place intentionally.

This project treats the visual problem seriously, but argues that solving it is only half the work. A nutrition system that answers every query with equal confidence, that has no mechanism for saying “this is not food” or “I am not sure between these three dishes”, and that cannot show where its numbers came from, is not safe to put in front of a user regardless of its headline accuracy. The proposed system is therefore organised around accuracy, reliability, explainability and grounding as four co-equal objectives.

Background

2.1 Transfer learning and modern vision backbones

The dominant paradigm in image classification is to adapt a model pretrained on a very large corpus rather than to train from scratch. Three families of pretraining are relevant here, and they fail in different ways, which is exactly why combining them is productive:

- **Language-supervised contrastive models.** SigLIP is trained to align images with their captions using a pairwise sigmoid loss. Because captions name dishes, the resulting features carry strong semantic priors about food categories.
- **Self-supervised models.** DINOv2 is trained without labels or text through self-distillation. Its features are sensitive to texture, surface and part structure — properties that distinguish a seared crust from a braised one — and are statistically decorrelated from contrastive features.
- **Supervised fine-grained models.** EVA-02 combines masked image modelling with supervised fine-tuning on ImageNet-22k, giving it a strong prior for exactly the kind of fine-grained discrimination Food-101 demands.

Because these three make different mistakes, an ensemble over them recovers accuracy that no single backbone achieves alone.

2.2 Frozen features as an experimental strategy

Fine-tuning a 300M-parameter backbone on consumer hardware costs hours per experiment, which in practice limits a student project to a handful of configurations chosen largely by guesswork. Running each backbone over the corpus exactly once and caching the pooled embeddings converts every subsequent experiment — head architecture, calibration method, conformal score, out-of-distribution rule, ensemble weighting — into a computation that finishes in seconds. The cache is what makes a rigorous, ablation-driven study feasible within the constraints of this course.

2.3 Calibration, conformal prediction and OOD detection

Modern neural networks are systematically overconfident: a softmax output of 0.95 does not correspond to being correct 95% of the time. Temperature scaling corrects this with a single parameter fitted on held-out data. Conformal prediction goes further

and returns a *set* of labels guaranteed, under an exchangeability assumption, to contain the true label at a chosen rate — so the system can honestly report “this is one of these three dishes” instead of guessing. Separately, any classifier restricted to 101 categories will confidently assign one of them to a photograph of a keyboard; an out-of-distribution detector is required for the system to decline.

2.4 Retrieval-augmented generation

RAG grounds a language model’s output in retrieved documents rather than its parameters. A production-grade pipeline requires considerably more than a single vector search: sparse and dense retrieval capture different query intents and must be fused; a cross-encoder reranker is needed because bi-encoder similarity is a weak relevance signal; corrective retrieval detects and repairs low-quality retrievals before generation; and grounding verification checks that generated claims are actually supported by the retrieved evidence. For a nutrition assistant, where a hallucinated figure is a safety issue, these are requirements rather than refinements.

Problem Statement

Existing photograph-based food recognition applications share a set of structural weaknesses:

1. **They emit point predictions without honest uncertainty.** A single label is returned with an uncalibrated confidence score. The user cannot distinguish a near-certain identification from a coin flip between two visually similar dishes.
2. **They cannot decline.** Presented with a photograph that contains no food, a 101-way classifier still returns one of its 101 categories, often with high confidence.
3. **Their nutritional figures are unsourced.** Numbers appear without provenance, and where a language model is involved they may be fabricated outright. In a health context this is not a cosmetic problem.
4. **They are opaque.** No indication is given of which region of the image drove the decision, leaving the user with no means of noticing an error.
5. **Portion size is ignored.** Nutritional data is published per 100 g, so a per-100 g figure presented as a per-serving figure can be wrong by a factor of three or more.

A preliminary experiment conducted during the preparation of this proposal illustrates how severe the grounding problem is in practice. Matching the 101 Food-101 category names against USDA SR Legacy by naive text search produced confidently incorrect results for a large fraction of classes: *beef carpaccio* resolved to “Soup, stock, beef” at 13 kcal per 100 g, *chicken wings* to “Soup, stock, chicken” at 36 kcal, *filet mignon* to a salmon product, and *chocolate cake* to a snack cake explicitly labelled “not chocolate”. Roughly a quarter of the categories matched no meaningful record at all. Any system that adopts such a mapping without curation will present users with calorie figures that are wrong by an order of magnitude while appearing entirely authoritative. This finding directly motivates the curated, auditable knowledge base described in Chapter 5.

This project therefore addresses the following question: *can a food recognition system be built on consumer hardware that is simultaneously highly accurate, statistically calibrated, capable of abstaining, visually explainable, and grounded in an auditable nutritional source?*

Objectives

1. Build a cached multi-backbone feature bank over Food-101 to make systematic experimentation computationally feasible.
2. Achieve 96% top-1 accuracy on the Food-101 test split through fine-tuning and ensembling, measured once under a protocol that prevents optimistic bias.
3. Calibrate predicted confidence and quantify the improvement with expected calibration error.
4. Produce conformal prediction sets with an empirically verified coverage guarantee.
5. Implement an out-of-distribution detector that rejects non-food imagery at a measured false-positive rate.
6. Generate visual explanations for every prediction, and estimate portion mass from monocular depth.
7. Construct a curated, fully auditable nutritional knowledge base covering all 101 categories, sourced from USDA SR Legacy.
8. Implement an advanced retrieval pipeline with hybrid search, reciprocal rank fusion, reranking, corrective retrieval, grounding verification and a graph-structured knowledge representation.
9. Evaluate that pipeline against a hand-curated gold-standard question set using standard retrieval and generation metrics.
10. Deliver the system as a deployed, publicly accessible web application with published model artefacts and reproducible training notebooks.

Proposed Methodology

The work is organised into four phases and fourteen stages.

5.1 Phase A — Perception

Stage 1: Frozen feature bank. Each of the three backbones is run exactly once over all 101,000 images and the pooled embeddings are cached to disk as half-precision arrays.

Stage 2: Classification heads. Probe classifiers are trained on the cached features. Because the embedding spaces of the three backbones are scaled differently, a *gated fusion* head is used: each backbone is projected to a common width and a learned gate weighs them per sample. The gate values are themselves interpretable, showing which backbone the model relies on for which foods.

Stage 3: Fine-tuning. EVA-02-L is fully fine-tuned at 448×448 with layer-wise learning-rate decay, mixup and CutMix, label smoothing, stochastic depth, cosine scheduling with warmup, and an exponential moving average of weights.

Stage 4: Ensemble and evaluation. The fine-tuned model and the fusion head are combined with test-time augmentation, and the result is evaluated once on the untouched test split.

5.2 Phase B — Reliability

Stage 5: Calibration, conformal prediction and OOD detection. Temperature and vector scaling are fitted on the validation split and assessed by expected calibration error. Adaptive prediction sets are constructed to achieve a target coverage level. An out-of-distribution detector combining energy-based scoring with Mahalanobis distance in feature space rejects non-food images.

Stage 6: Explainability and portion estimation. Grad-CAM and attention rollout produce per-prediction visual explanations. Monocular depth estimation combined with the segmented food region yields a volume proxy, converted to mass using per-category density priors.

5.3 Phase C — Knowledge and retrieval

Stage 7: Nutrition knowledge base. A curated mapping is built from each of the 101 categories to USDA SR Legacy. Where SR Legacy contains the dish, a specific record is pinned; where it does not — roughly thirty categories, including tiramisu, bibimbap and takoyaki — the dish is represented as a weighted composite of ingredient

records, which additionally supplies the dish-to-ingredient edges used in Stage 8. Every entry is auditable back to its source records.

Stage 8: Advanced RAG pipeline. Sparse BM25 and dense embedding retrieval are fused by reciprocal rank fusion, reranked by a cross-encoder, checked and repaired by corrective retrieval, verified for grounding, and augmented by a graph representation linking dishes, ingredients and nutrients. Generation is performed by an OpenAI model behind a provider-agnostic interface; all retrieval components run locally.

Stage 9: RAG evaluation. A hand-curated gold-standard question set is used to measure retrieval quality (recall@ k , MRR, nDCG) and generation quality (faithfulness, answer relevance, context precision and recall).

5.4 Phase D — Delivery

Stages 10–14. Models are exported to ONNX and published to the Hugging Face Hub; a FastAPI backend is containerised for Hugging Face Spaces; a Next.js front end is built on a bespoke comic-inspired design system; training notebooks, architecture documentation, an illustrated README, a written report and slides are produced; and the complete system is deployed.

System Architecture

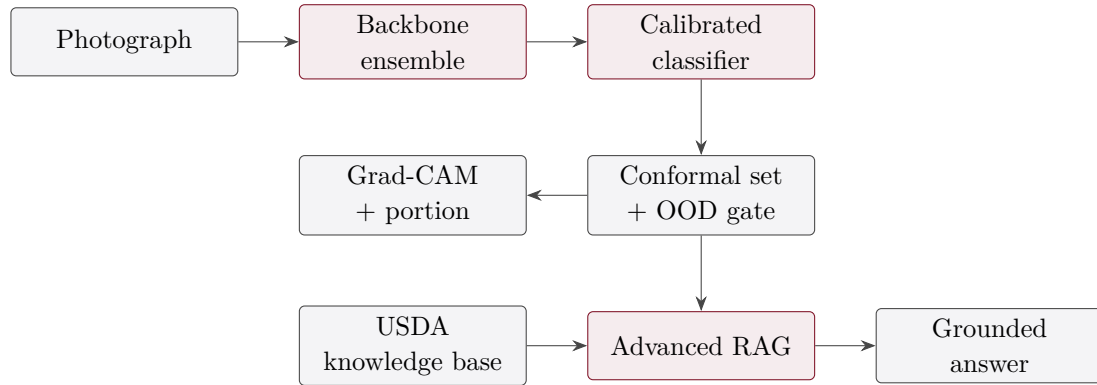


Figure 6.1: End-to-end flow. A photograph is embedded by the backbone ensemble, classified by a calibrated head, gated by conformal and out-of-distribution checks, explained visually, and finally answered against the curated knowledge base through the retrieval pipeline.

Dataset and Experimental Setup

7.1 Food-101

Food-101 contains 101 categories with 1,000 images each. The official split provides 750 training and 250 test images per class.

Property	Value
Categories	101
Training images	75,750
Test images	25,250
Total images	101,000
On-disk size	6.1 GB

Table 7.1: Food-101 as downloaded and verified locally.

Food-101 ships only a training and a test split. Reporting model-selection results on the test split is the single most common way projects of this kind overstate their accuracy. A fixed, class-stratified 4% validation slice is therefore carved out of the training split under a fixed random seed and shared by both the image and the cached-feature pipelines, and **the test split is not consulted until final evaluation**.

7.2 USDA SR Legacy

Nutritional data is taken from USDA FoodData Central’s SR Legacy release, the last USDA release providing complete micronutrient profiles for generic, unbranded foods — the correct reference when the input is a dish rather than a packaged product. Thirty-two nutrients per food are tracked, along with published portion weights.

7.3 Hardware and measured throughput

All training is performed locally on an Apple M3 Pro with 18 GB of unified memory using PyTorch’s Metal backend. The following throughput figures were measured directly rather than estimated.

Backbone	Res.	Params	Throughput	Full corpus
SigLIP-SO400M	384	428.2M	5.3 img/s	~316 min
EVA-02-L	448	304.1M	4.1 img/s	~414 min
DINOv2-L	518	304.4M	3.6 img/s	~464 min

Table 7.2: Measured inference throughput at batch size 8 in half precision. The one-off cost of building the feature bank is approximately 20 hours; every experiment afterwards runs against the cache in seconds.

Evaluation Plan

Dimension	Metrics
Classification	Top-1 and top-5 accuracy, macro-F1, per-class accuracy, confusion analysis of the hardest pairs
Calibration	Expected and maximum calibration error, reliability diagrams, Brier score
Conformal prediction	Empirical coverage against the target level, average prediction-set size
OOD detection	AUROC and FPR at 95% true-positive rate against non-food imagery
Portion estimation	Mean absolute percentage error against a hand-measured reference set
Retrieval	Recall@ k , MRR, nDCG on the gold-standard question set
Generation	Faithfulness, answer relevance, context precision and recall
System	End-to-end latency, model size, cold-start time

Table 8.1: Evaluation dimensions and the metrics reported for each.

Reported accuracy will be compared against published Food-101 results to establish where the system sits relative to the literature. All numbers will be produced by scripts committed to the repository so that they can be regenerated.

Expected Outcomes

- A Food-101 classifier reaching approximately 96% top-1 accuracy, measured under a protocol that prevents test-set leakage.
- Calibrated confidence estimates and conformal prediction sets with a verified coverage guarantee.
- A working out-of-distribution rejector for non-food photographs.
- Visual explanations and depth-based portion estimates for every prediction.
- A curated, auditable nutritional knowledge base covering all 101 categories.
- An advanced retrieval pipeline with a measured evaluation against a hand-built gold standard.
- Public artefacts: models on the Hugging Face Hub, a containerised API on Hugging Face Spaces, a deployed web application, and a documented open-source repository with notebooks covering every training stage.

Tools and Technologies

Area	Technology
Language	Python 3.11
Deep learning	PyTorch 2.13 (Metal backend), timm, Hugging Face Transformers
Data	Hugging Face Datasets, NumPy, pandas, Pillow
Retrieval	sentence-transformers, FAISS, BM25, cross-encoder reranking
Generation	OpenAI API behind a provider-agnostic interface
Backend	FastAPI, ONNX Runtime, Docker, Hugging Face Spaces
Frontend	Next.js, TypeScript, Vercel
Reporting	Jupyter, Matplotlib, L ^A T _E X, Mermaid

Timeline

Week	Phase	Deliverable
1	A	Feature bank complete; probe heads and fusion ensemble trained
2	A	EVA-02-L fine-tuned at 448; ensemble with TTA; final test evaluation
3	B	Calibration, conformal prediction, OOD detection, Grad-CAM, portion estimation
4	C	Curated nutrition knowledge base for all 101 categories
5	C	Advanced RAG pipeline and its gold-standard evaluation
6	D	ONNX export, Hub publication, FastAPI backend, Docker image
7	D	Next.js front end and deployment
8	D	Notebooks, documentation, report and slides

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